



JABchem



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Past Papers Higher Chemistry

2023 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/120)	%	
A	80+	66.7%	32.6%
B	65+	54.1%	23.9%
C	50+	41.7%	21.3%
D	35+	29.2%	14.4%
No award	<35	<29.2%	7.8%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	17.6 /25	49.7 /95	No Assignment in 2023

2023 Higher Chemistry Marking Scheme

MC Qu	Answer	% Correct	Reasoning
1	A	71	☒ A Electronegativities Na = 0.9 & I = 2.7 ∴ difference = 1.8 (least ionic character) ☒ B Electronegativities Na = 0.9 & F = 4.0 ∴ difference = 3.1 ☒ C Electronegativities K = 0.8 & I = 2.7 ∴ difference = 1.9 ☒ D Electronegativities K = 0.8 & F = 4.0 ∴ difference = 3.2 (greatest ionic character)
2	D	89	☒ A N-H bond in structure ∴ hydrogen bonding would occur between molecules ☒ B O-H bond in structure ∴ hydrogen bonding would occur between molecules ☒ C N-H bond in structure ∴ hydrogen bonding would occur between molecules ☒ D No N-H, O-H or H-F bonds in structure ∴ no hydrogen bonding between molecules
3	C	85	
4	A	69	☒ A Activation of reverse reaction = $200 - 50 = +150 \text{ kJ mol}^{-1}$ ☒ B Enthalpy change for reverse reaction = $50 - 150 = -100 \text{ kJ mol}^{-1}$ ☒ C Activation of forward reaction = $200 - 150 = 150 \text{ kJ mol}^{-1}$ ☒ D Enthalpy change for forward reaction = $150 - 50 = +100 \text{ kJ mol}^{-1}$
5	C	91	$\text{time} = 10\text{s} \quad \therefore \text{rate} = \frac{1}{\text{time}} = \frac{1}{10\text{s}} = 0.1 \text{ s}^{-1}$ Extrapolate from graph: when rate = 0.1 s^{-1} then concentration = 0.25 mol l^{-1}
6	B	74	☒ A catalyst speeds up reaction so dotted line would be steeper initially ☒ B dotted line is steeper at start and meets full line at horizontal end of line ☒ C catalyst does not change final concentrations so dotted line would meet full line at end ☒ D catalyst speeds up reaction so dotted line would be steeper initially
7	B	80	$1 \text{ mol CH}_3\text{OH} = -726 \text{ kJ mol}^{-1} = 32\text{g}$ $-145.2 \text{ kJ mol}^{-1} = 32\text{g} \times \frac{-145.2}{-726}$ $= 6.4\text{g}$
8	A	83	☒ A Sodium Na atom (covalent radius = 160pm) is larger than sodium Na^+ ion (ionic radius = 102pm) ☒ B Chloride Cl^- ion (ionic radius = 181pm) is larger than chlorine Cl atom (covalent radius = 100pm) ☒ C Magnesium Mg^{2+} ion (ionic radius = 72pm) is smaller than magnesium Mg atom (covalent radius = 140pm) ☒ D Oxygen O atom (covalent radius = 64pm) is smaller than oxide O^{2-} ion (ionic radius = 140pm)
9	C	87	☒ A Covalent molecular is found in both elements e.g. O_2 and compounds e.g. H_2O ☒ B Covalent network is found in both elements e.g. C (diamond) and compounds e.g. SiO_2 ☒ C Monoatomic structures are only found in elements ☒ D Ionic structures are only found in compounds containing metals and non-metal elements
10	B	72	☒ A 2-methylbutanal $\text{C}_5\text{H}_{10}\text{O}$ is <u>not</u> an isomer of hexanal $\text{C}_6\text{H}_{12}\text{O}$ ☒ B 3-methylpentan-2-one $\text{C}_6\text{H}_{12}\text{O}$ is an isomer of hexanal $\text{C}_6\text{H}_{12}\text{O}$ (same formula different structure) ☒ C 2,2-dimethylbutan-1-ol $\text{C}_6\text{H}_{14}\text{O}$ is <u>not</u> an isomer of hexanal $\text{C}_6\text{H}_{12}\text{O}$ ☒ D 3,3-dimethylpentanal $\text{C}_7\text{H}_{14}\text{O}$ is <u>not</u> an isomer of hexanal $\text{C}_6\text{H}_{12}\text{O}$

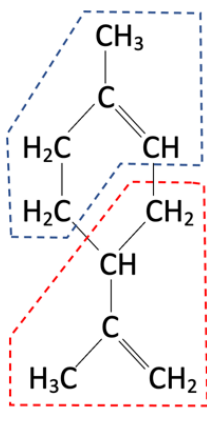
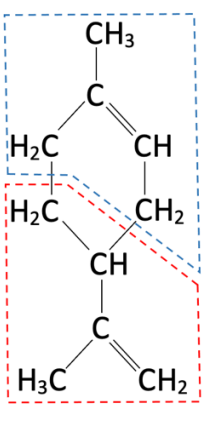
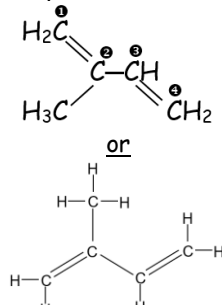
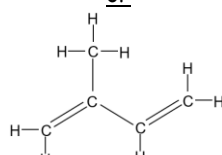
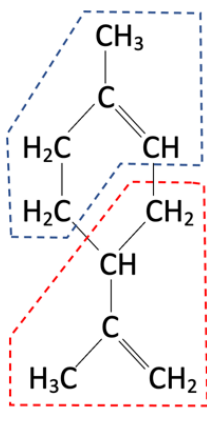
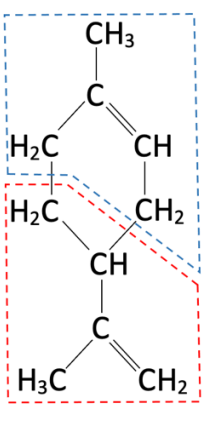
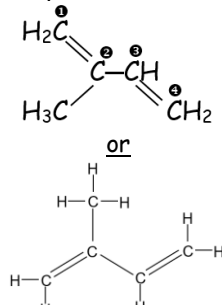
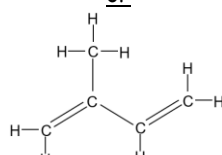
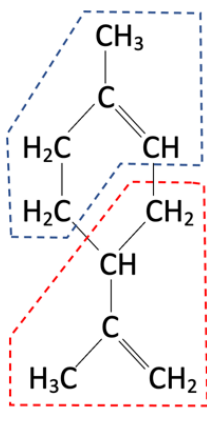
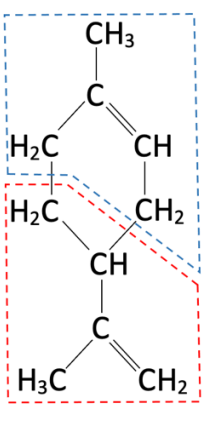
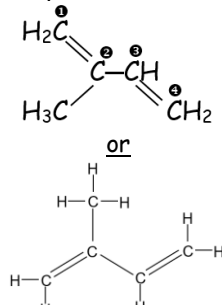
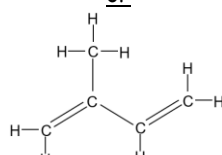
11	C	83	carboxyl group amine peptide link
12	D	50	☒ A Formation of carbon monoxide indicated incomplete combustion not complete combustion ☒ B Formation of carbon monoxide indicated incomplete combustion not complete combustion ☒ C enthalpy of combustion is the complete combustion of 1 mole of a substance ☒ D The energy change for the complete combustion of 1 mol of a substance
13	D	28	☒ A ethyl ethanoate and water in B will react to form ethanol and ethanoic acid and form an equilibrium ☒ B ethyl ethanoate and water in B will react to form ethanol and ethanoic acid and form an equilibrium ☒ C ethanol and ethanoic acid react to form ethyl ethanoate and water in A and form an equilibrium ☒ D ethyl ethanoate and water in B will react to form ethanol and ethanoic acid and form an equilibrium ☒ D Flask A is the reactants at 100% and Flask B is the products at 100%. They will reach the same equilibrium eventually with ethyl ethanoate, water, ethanol and ethanoic acid present in both flasks at the same concentrations.
14	C	70	Write Down main species IO_3^- $\longrightarrow$ I_2 Balance all atoms other than oxygen and hydrogen 2IO_3^- $\longrightarrow$ I_2 Balance O by adding H_2O to other side 2IO_3^- $\longrightarrow$ $\text{I}_2 + 6\text{H}_2\text{O}$ Balance H by adding H^+ ions to other side $2\text{IO}_3^- + 12\text{H}^+$ $\longrightarrow$ $\text{I}_2 + 6\text{H}_2\text{O}$ Balance charge by adding e^- to most positive side $2\text{IO}_3^- + 12\text{H}^+ + 10e^-$ $\longrightarrow$ $\text{I}_2 + 6\text{H}_2\text{O}$
15	D	78	☒ A Condensation: 2 molecules join together with small molecule e.g. water removed at join ☒ B Hydration: Addition reaction where H_2O is added across a $\text{C}=\text{C}$ double bond ☒ C Reduction: decrease in oxygen : hydrogen ratio ☒ D Oxidation: increase in oxygen : hydrogen ratio ($\text{C}_6\text{H}_7\text{ON} \rightarrow \text{C}_6\text{H}_5\text{O}_2\text{N}$) O:H ratio 1:7 $\rightarrow$ 2:5
16	B	46	☒ A 3x -OH group makes A the 2 nd most polar molecule and A is the 3 rd peak left to right ☒ B no -OH groups makes B the least polar molecule and B is the 1 st peak left to right (Z) ☒ C 2x -OH group makes C the 3 rd most polar molecule and C is the 2 nd peak left to right ☒ D 6x -OH group makes D the most polar molecule and D is the 4 th peak left to right
17	C	59	☒ A Draft shield would reduce heat loss to the surroundings ☒ B the thermometer in the diagram is too close to the flame and might give inaccurate temp ☒ C glass beaker would reduce the heat transfer to the water compared to a copper can ☒ D Stirring the water would mix the water better and give a more accurate temp
18	B	85	☒ A head of soap is ionic and therefore polar ☒ B ionic head dissolves in polar water and non-polar tail dissolves in non-polar oil ☒ C head of soap is ionic and therefore polar ☒ D ionic head is polar and dissolves in polar water rather than non-polar head
19	C	53	☒ A unreacted nickel oxide must be removed by filtration before evaporation takes place ☒ B the filtration of nickel oxide must be followed by evaporation of water to form salt ☒ C the unreacted nickel oxide is filtered to remove it and the evaporation that follows removes the water from the nickel sulfate solution to leave nickel sulphate salt ☒ D the unreacted nickel oxide must be filtered before evaporation takes place
20	B	76	☒ A ethanal CH_3CHO is an aldehyde and does not react with alkalis ☒ B ethanoic acid CH_3COOH is a carboxylic acid and is neutralised by alkalis. ☒ C propanone CH_3COCH_3 is a ketone and does not react with alkalis ☒ D ethan-1-ol $\text{CH}_3\text{CH}_2\text{OH}$ is a primary alcohol and does not react with alkalis

21	D	94	☒ A Secondary alcohol: 2 carbons directly attached to the carbon with -OH group ☒ B Secondary alcohol: 2 carbons directly attached to the carbon with -OH group ☒ C Tertiary alcohol: 3 carbons directly attached to the carbon with -OH group ☒ D Primary alcohol: 1 carbon directly attached to the carbon with -OH group		
22	A	68	4-methylpentan-2-one is a ketone which reduces to form the secondary alcohol 4-methylpentan-2-ol. gfm 4-methylpentan-2-one $C_6H_{12}O = (6 \times 12) + (12 \times 1) + (1 \times 16) = 72 + 12 + 16 = 100g$ gfm 4-methylpentan-2-ol $C_6H_{14}O = (6 \times 12) + (14 \times 1) + (1 \times 16) = 72 + 14 + 16 = 102g$ 4-methylpentan-2-one $C_6H_{12}O$ 1mol 100g → 4-methylpentan-2-ol $C_6H_{14}O$ 1mol 102g		
23	D	42	gfm $O_2 = 32g \therefore$ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{16}{32} = 0.5\text{mol}$ (same no. of mol = same volume) ☒ A gfm $CO = 28g \therefore$ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{21}{28} = 0.75\text{mol}$ ☒ B gfm $CO_2 = 44g \therefore$ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{44}{44} = 1.0\text{mol}$ ☒ C gfm $NO_2 = 46g \therefore$ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{46}{46} = 1.0\text{mol}$ ☒ D gfm $N_2O_4 = 92g \therefore$ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{46}{92} = 0.5\text{mol}$		
24	A	45	Ionic formula aluminium sulfate = $(Al^{3+})_2(SO_4^{2-})_3$ 1 mol of $Al_2(SO_4)_3$ contains 2 mol of positive Al^{3+} ions 0.25mol 0.5 mol		
25	B	56	<table style="width: 100%; border-collapse: collapse;"><tr><td style="width: 50%; padding: 5px;">C_2 in $C=C$ double bond has no hydrogens directly attached to it $\therefore$ Cl of $H-Cl$ attaches to C_2 (major product)</td><td style="width: 50%; padding: 5px;">C_3 in $C=C$ double bond has 1 hydrogen directly attached to it $\therefore$ H of $H-Cl$ attaches to C_3 (major product)</td></tr></table> 2-methylpent-2-ene $\begin{array}{ccccccc} & H & CH_3 & & H & H & \\ & & & & & & \\ H - & C^1 - & C^2 = & C^3 - & C^4 - & C^5 - & H \\ & & & & & & \\ & H & & H & H & H & \end{array}$ → 2-chloro-2-methylpentane $\begin{array}{ccccccc} & H & CH_3 & H & H & H & \\ & & & & & & \\ H - & C - & C - & C - & C - & C - & H \\ & & & & & & \\ & H & Cl & H & H & H & \end{array}$ + Cl - H	C_2 in $C=C$ double bond has no hydrogens directly attached to it $\therefore$ Cl of $H-Cl$ attaches to C_2 (major product)	C_3 in $C=C$ double bond has 1 hydrogen directly attached to it $\therefore$ H of $H-Cl$ attaches to C_3 (major product)
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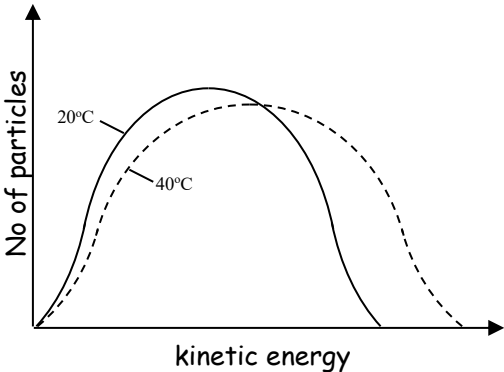
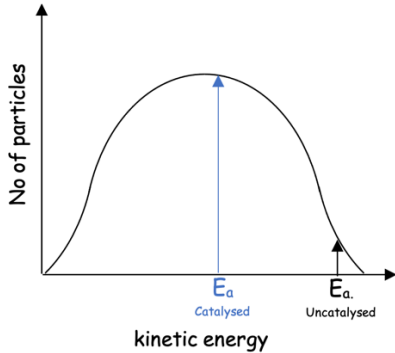
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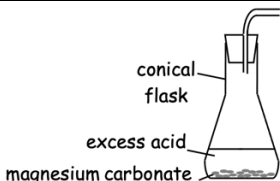
Long Qu	Answer	Reasoning																																
1a(i)	Greater no. of protons/ nuclear charge holds electrons more tightly	Across a period, the same electron shell is filling up but there is a greater nuclear charge due to the increase atomic number. The outer shell is held more tightly by the nucleus and an electron is harder to remove from the outer shell.																																
1a(ii)	b or j	Elements c and k are group 0 elements as they have the highest 1 st ionisation energy in their periods. Group 7 elements have the next highest 1 st ionisation energy and elements b and j correspond to the 2 nd highest ionisation energies.																																
1a(iii)A	Answer to include:	<table><tr><td>1st mark:</td><td>2nd electron removed from an electron shell closer to nucleus</td></tr><tr><td>2nd mark:</td><td>2nd electron is less screened/shielded (than 1st electron removed) or 2nd electron is more strongly attracted to the nucleus</td></tr></table>	1 st mark:	2 nd electron removed from an electron shell closer to nucleus	2 nd mark:	2 nd electron is less screened/shielded (than 1 st electron removed) or 2 nd electron is more strongly attracted to the nucleus																												
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1a(iii)B	11472	<table><tr><td>①</td><td>Na⁺_(g)</td><td>→</td><td>Na²⁺_(g) + e⁻</td><td>ΔH = 4562 kJ mol⁻¹</td></tr><tr><td>②</td><td>Na²⁺_(g)</td><td>→</td><td>Na³⁺_(g) + e⁻</td><td>ΔH = 6910 kJ mol⁻¹</td></tr><tr><td>add</td><td>Na⁺_(g)</td><td>→</td><td>Na³⁺_(g) + 2e⁻</td><td>ΔH = 11472kJ mol⁻¹</td></tr></table>	①	Na ⁺ _(g)	→	Na ²⁺ _(g) + e ⁻	ΔH = 4562 kJ mol ⁻¹	②	Na ²⁺ _(g)	→	Na ³⁺ _(g) + e ⁻	ΔH = 6910 kJ mol ⁻¹	add	Na ⁺ _(g)	→	Na ³⁺ _(g) + 2e ⁻	ΔH = 11472kJ mol ⁻¹																	
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1b(i)	Attraction atom/nucleus has for electrons within a bond	Electronegativity is a measure of the attraction an atom involved in a bond has for the electrons of the bond (shared pair of electrons)																																
1b(ii)	Answer to include:	<table><tr><td>1st mark:</td><td>Increased screening/shielding effect (due to more shells)</td></tr><tr><td>2nd mark:</td><td>Attraction of nucleus/protons for outer electrons decreases</td></tr></table>	1 st mark:	Increased screening/shielding effect (due to more shells)	2 nd mark:	Attraction of nucleus/protons for outer electrons decreases																												
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1b(iii)	Strontium (or Barium or Radium)	Strongest reducing agents are found on the top right hand corner of the electrochemical series. First four in top right corner are group 1 elements and Strontium is the first group 2 element.																																
2a(i)	Answer to Include:	<table><tr><td>1st mark:</td><td>(intermolecular) forces increase going down a group</td></tr><tr><td>2nd mark:</td><td>London dispersion forces are forces broken between molecules</td></tr><tr><td>3rd mark:</td><td>The more electrons the stronger the LDFs</td></tr></table>	1 st mark:	(intermolecular) forces increase going down a group	2 nd mark:	London dispersion forces are forces broken between molecules	3 rd mark:	The more electrons the stronger the LDFs																										
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2a(ii)	Hydrogen bonding	Hydrogen bonding occurs between molecules containing one of the following bonds: <table><tr><td>N - H bond</td><td>O - H bond</td><td>H - F bond</td></tr></table>	N - H bond	O - H bond	H - F bond																													
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2b(i)	34	<table><tr><td>Desired equation</td><td colspan="3"> Si + 2H₂ → SiH₄ </td></tr><tr><td>①</td><td>SiH₄ + 2O₂ → SiO₂ + 2H₂O</td><td>ΔH=-1517 kJ mol⁻¹</td><td></td></tr><tr><td>②</td><td>Si + O₂ → SiO₂</td><td>ΔH= -911 kJ mol⁻¹</td><td></td></tr><tr><td>③</td><td>H₂ + ½O₂ → H₂O</td><td>ΔH= -286 kJ mol⁻¹</td><td></td></tr><tr><td>①x-1</td><td>SiO₂ + 2H₂O → SiH₄ + 2O₂</td><td>ΔH=+1517 kJ mol⁻¹</td><td></td></tr><tr><td>②</td><td>Si + O₂ → SiO₂</td><td>ΔH= -911 kJ mol⁻¹</td><td></td></tr><tr><td>③x2</td><td>2H₂ + O₂ → 2H₂O</td><td>ΔH= -572 kJ mol⁻¹</td><td></td></tr><tr><td>add</td><td>Si + 2H₂ → SiH₄</td><td>ΔH= +34 kJ mol⁻¹</td><td></td></tr></table>	Desired equation	Si + 2H₂ → SiH₄			①	SiH ₄ + 2O ₂ → SiO ₂ + 2H ₂ O	ΔH=-1517 kJ mol ⁻¹		②	Si + O ₂ → SiO ₂	ΔH= -911 kJ mol ⁻¹		③	H ₂ + ½O ₂ → H ₂ O	ΔH= -286 kJ mol ⁻¹		①x-1	SiO₂ + 2H₂O → SiH₄ + 2O₂	ΔH=+1517 kJ mol⁻¹		②	Si + O₂ → SiO₂	ΔH= -911 kJ mol⁻¹		③x2	2H₂ + O₂ → 2H₂O	ΔH= -572 kJ mol⁻¹		add	Si + 2H ₂ → SiH ₄	ΔH= +34 kJ mol ⁻¹	
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2b(ii)	40%	gfm Mg₂Si = 76.7g v no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{15.32}{76.7} = 0.200\text{mol}$ 4HCl + Mg₂Si → SiH₄+ 2MgCl₂ 1mol																																

2b(iii)	Answer to include:	1 st mark:	In SiO ₂ covalent bonds are broken		
		2 nd mark:	In SiH ₄	London dispersion forces Van der Waals' forces Intermolecular forced	are broken
		3 rd mark:	Covalent bonds	are stronger need more energy to break	than London dispersion forces Van der Waals' forces Intermolecular forced
3a(i)A	propane-1,2,3-triol	H H H H—C—C—C—H OH OH OH glycerol propane-1,2,3-triol 3 carbons Single bonds between carbons Functional groups on Carbons 1,2,3 3 hydroxyl -OH groups			
3a(i)B	condensation	A condensation reaction occurs when two molecules join together with a small molecule removed where they join. The small molecule is usually water.			
3a(ii)A	carbonyl	O —C—	O —C—OH	O —C—H	O C—C—C
		carbonyl group	carboxyl group	aldehyde group	ketone group
3a(ii)B	heptan-2-one	Seven carbons in main chain. (hept) Carbonyl functional group between 2 carbons = Ketone (heptanone) Functional group in C ₂ (heptan-2-one)			
3a(ii)C	reduction	Primary Alcohol	Secondary Alcohol	Carboxylic Acid	
		Aldehyde	Ketone	Aldehyde	Ketone
		Carboxylic Acid		Primary Alcohol	Secondary Alcohol
3a(ii)D	Reaction 1	① C₈H₁₆O₂ O : H 2 : 16 1 : 8 → ② C₈H₁₄O₃ O : H 3 : 14 1 : 4.67 → ③ C₇H₁₄O O : H 1 : 14 1 : 14 → C₇H₁₆O O : H 1 : 16 1 : 16 → → → →			

3b(iii)	denaturing	When a protein changes shape by a change like heating, the protein changes shape as hydrogen bonds are broken to alter the shape of the protein e.g. spirals of protein unravel.														
3c	To prevent non-polar and polar liquids separating (into layers)	Emulsifiers had hydrophilic polar. Heads and hydrophobic non-polar tails. The hydrophobic tails stick into non-polar oil/grease droplets and hydrophilic heads stay outside the oil/grease droplets in the polar water. The oil/grease droplets cannot join together are form an oil/grease layer which spoils the appearance of the food.														
3d	Calcium propanoate	$\text{Ca}^{2+}(\text{CH}_3\text{CH}_2\text{COO}^-)_2$ is the salt formed by the neutralisation of propanoic acid ($\text{CH}_3\text{CH}_2\text{COOH}$) with a calcium-containing base e.g. calcium hydroxide														
4	Open Ended Question:	<table><tr><th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr><tr><td>Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>			3 mark answer	2 mark answer	1 mark answer	Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.						
		3 mark answer	2 mark answer	1 mark answer												
Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.														
<table><tr><th colspan="4">Candidate Marks</th></tr><tr><td>0 marks</td><td>1 mark</td><td>2 marks</td><td>3 marks</td></tr><tr><td>31%</td><td>33%</td><td>23%</td><td>13%</td></tr></table>		Candidate Marks				0 marks	1 mark	2 marks	3 marks	31%	33%	23%	13%			
Candidate Marks																
0 marks	1 mark	2 marks	3 marks													
31%	33%	23%	13%													
5a(i)A	494	0.05litres glucose solution contains 5.79g glucose 16litres glucose solution contains 5.79g glucose $\times \frac{16}{0.05}$ = 1852.8g glucose gfm $\text{C}_6\text{H}_{12}\text{O}_6$ = 180g $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{1852.8}{180} = 10.3\text{mol}$ $\text{C}_6\text{H}_{12}\text{O}_6 \longrightarrow 2\text{C}_2\text{H}_5\text{OH} + 2\text{CO}_2$ <table><tr><td>1mol</td><td>2mol</td><td>2mol</td></tr><tr><td>10.3mol</td><td></td><td>20.6mol</td></tr></table> Volume = no. of mol $\times$ Molar Volume = 20.6mol $\times$ 24litres mol^{-1} = 494litres			1mol	2mol	2mol	10.3mol		20.6mol						
1mol	2mol	2mol														
10.3mol		20.6mol														
5a(i)B	51.1	atom economy = $\frac{\text{Mass of useful produce}}{\text{Mass of reactants}} \times 100 = \frac{2 \times 46}{1 \times 180} \times 100 = \frac{92}{180} \times 100 = 51.1\%$														
5a(ii)	12.2	% alcohol by volume = $\frac{\text{Change in specific gravity}}{0.7362} \times 100 = \frac{1.075 - 0.985}{0.7362} \times 100 = 12.2\%$														
5b(i)	Answer to include one from:	<table><tr><td>Acidified dichromate would turn orange to green with methanol and no colour change with propan-2-one.</td><td>Hot copper (II) oxide would turn black to brown with methanol and no colour change with propan-2-one</td></tr></table> The oxidising agents Fehling's solution and Tollen's reagent would not react with methanol or propan-2-one			Acidified dichromate would turn orange to green with methanol and no colour change with propan-2-one.	Hot copper (II) oxide would turn black to brown with methanol and no colour change with propan-2-one										
Acidified dichromate would turn orange to green with methanol and no colour change with propan-2-one.	Hot copper (II) oxide would turn black to brown with methanol and no colour change with propan-2-one															
5b(ii)A	terpenes	Terpenes are formed when isoprene C_5H_8 units join together. The terpenes formed have a carbon number with is a multiple of 5, depending in how many isoprenes joined together.														
5b(ii)B	C_5H_8 unit circled as shown opposite:	<table><tr><td></td><td></td><td colspan="2">Circled area should resemble a five carbon isoprene unit shown below  or  • C=C double bonds alter as units join together • start at one end of terpene to find a C_5H_8 unit </td></tr></table>					Circled area should resemble a five carbon isoprene unit shown below  or  • C=C double bonds alter as units join together • start at one end of terpene to find a C_5H_8 unit									
		Circled area should resemble a five carbon isoprene unit shown below  or  • C=C double bonds alter as units join together • start at one end of terpene to find a C_5H_8 unit														

5b(ii)C	ethanoic acid	ethanoic acid generiol
5c	2100mg or 2.1g (1 mark for mass and 1 mark for unit)	For 8 hours: 1kg body weight = 10.0mg quinine 70kg body weight = 10.0mg quinine $\times \frac{70}{1}$ = 700mg quinine For 24 hours: = 700mg $\times \frac{24}{8}$ = 2100mg (=2.1g)
6a(i)A	Curve finishes below reactants	Exothermic
6a(i)B	Activated Complex	reactants activated complex products
6a(ii)	Calculation showing:	no. of mol HNO_3 = volume $\times$ concentration = 1316 litres $\times$ 9.5 mol l^{-1} = 12502 mol no. of mol NH_3 = $\frac{\text{mass}}{\text{gfm}}$ = $\frac{220000}{17}$ = 12941 mol (available) $\text{HNO}_3 + \text{NH}_3 \longrightarrow \text{NH}_4\text{NO}_3$ 1mol 1mol 12502mol 12502mol(required) More NH_3 available (12941mol) than is required (12502mol) to react all HNO_3 $\therefore \text{NH}_3$ is in excess and HNO_3 is the limiting reactant
6a(iii)	100% atom economy	As there is only one product then all reactants end up in the useful product.
6b(i)A	Total number of particles/molecules	The area under the curve is the total number of particles in the sample.

6b(i)B	Graph moves to right			
6b(ii)	Line to the left of dotted line diagram			
6c(i)	rate of forward reaction equals rate of reverse reaction	At equilibrium: rate of forward reaction = rate of reverse reaction (and concentration of reactants and products are constant)		
6c(ii)A	Answer to include:	1 st mark:	equilibrium shifts to right hand side or increases yield of ammonia	
		2 nd mark:	decreases the rate of reverse reaction or rate of forward reaction is then greater than rate of reverse reaction	
6c(ii)B	One answer from:	recycling of unused gases/reactants	air is a low cost or free resource	uses a catalyst (to reduce energy costs)
6d(i)	$4\text{NH}_3 + 3\text{O}_2 \rightarrow 2\text{N}_2 + 6\text{H}_2\text{O}$	① $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$ ② $2\text{NH}_3 + 6\text{OH}^- \rightarrow \text{N}_2 + 6\text{H}_2\text{O} + 6\text{e}^-$ ①x3 $3\text{O}_2 + 6\text{H}_2\text{O} + 12\text{e}^- \rightarrow 12\text{OH}^-$ ②x2 $4\text{NH}_3 + 12\text{OH}^- \rightarrow 2\text{N}_2 + 12\text{H}_2\text{O} + 12\text{e}^-$ add $3\text{O}_2 + \cancel{6\text{H}_2\text{O}} + \cancel{12\text{e}^-} \rightarrow \cancel{12\text{OH}^-} + 2\text{N}_2 + \cancel{12\text{H}_2\text{O}} + \cancel{12\text{e}^-}$ $3\text{O}_2 + 4\text{NH}_3 \rightarrow 2\text{N}_2 + 6\text{H}_2\text{O}$		
6d(ii)	ammonia or NH ₃	Reducing Agent: reduced something else and is oxidised itself. NH ₃ is oxidised and loses electrons: $2\text{NH}_3 + 6\text{OH}^- \rightarrow \text{N}_2 + 6\text{H}_2\text{O} + 6\text{e}^-$		
7a(i)	hard water	Hard water contains large quantities of (usually) calcium Ca ²⁺ ions. These ions react with soap to form an insoluble precipitate known as scum.		
7a(ii)	hydrophilic	hydrophilic	The ionic head of a detergent molecule is hydrophilic as it is polar and dissolved in water	
		hydrophobic	The hydrocarbon tail of a detergent molecule is hydrophobic as it is non-polar and will not mix with water. The hydrocarbon tail will stick into non-polar oil/grease instead	
7a(iii)	8.37x10 ⁻⁴ or 0.000837	no. of mol EDTA = volume x concentration = 0.0093litres x 0.0045mol l ⁻¹ = 4.185x10 ⁻⁵ mol $\begin{array}{c} \text{Ca}^{2+} + \text{C}_{10}\text{H}_{12}\text{N}_2\text{O}_8^{4-} \longrightarrow [\text{Ca}(\text{C}_{10}\text{H}_{12}\text{N}_2\text{O}_4)]^{2-} \\ 1\text{mol} \quad 1\text{mol} \\ 4.185 \times 10^{-5} \text{ mol} \quad 4.185 \times 10^{-5} \text{ mol} \end{array}$ concentration = $\frac{\text{no. of mol}}{\text{volume}} = \frac{4.185 \times 10^{-5} \text{ mol}}{0.05 \text{ litres}} = 8.37 \times 10^{-4} \text{ mol}$		

10a	Answer to include:	Mark	Version 1	Version 2		Version 3
		1 st Mark	Tare the balance with the crucible	Tare a weighing boat, transfer the 1.5 g onto the weighing boat. Record mass. Transfer into the crucible.		Weigh mixture and weighing boat, record the mass. Transfer mixture into the crucible.
		2 nd Mark	Transfer 1.5 g (into the crucible)	Reweigh the weighing boat and record the mass/calculate the difference		Reweigh the weighing boat and record the mass/calculate the difference
10b(i)	to allow gas/ CO_2 to escape	The gas formed in the reaction will build up in the crucible under the lid. Lifting the lid will prevent. Pressure building up that could dislodge the lid.				
10b(ii)	reactants/products are not flammable	Bunsen burners should be replaced with non-flammable heating methods like hot plates and heating mantles when the reactants or products are flammable.				
10c	0.582	mass of crucible before heating = 1.598g mass of crucible after heating = 1.294g mass of CO_2 released = 1.598g - 1.294g = 0.304g gfm CO_2 = 44g $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{0.304}{44} = 0.00691\text{mol}$ MgCO_3 1mol 0.00691mol → $\text{MgO} + \text{CO}_2$ 1mol 0.00691mol gfm MgCO_3 = 84.3g mass = no. of mol × gfm = 0.00691 × 84.3 = 0.582g				
10d(i)						
10d(ii)	carbon dioxide has low solubility in water	Only gases that are insoluble or have low solubility in water should be collect over water this water. There is some loss of gas during the process. The best way to collect CO_2 in a gas syringe so there is no loss of any gas by dissolving in water.				

2023 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	A	71	7	10	12	0	0
2	D	3	3	5	89	0	0
3	C	4	3	85	8	0	0
4	A	69	13	13	5	0	0
5	C	2	5	91	2	0	0
6	B	5	74	7	13	0	0
7	B	5	80	4	11	1	0
8	A	83	5	5	6	0	0
9	C	3	5	87	5	0	0
10	B	6	72	17	5	0	0
11	C	9	3	83	5	0	0
12	D	14	20	15	50	1	0
13	D	10	31	31	28	0	0
14	C	6	8	70	16	0	0
15	D	11	5	6	78	0	0
16	B	36	46	9	9	0	0
17	C	6	13	59	22	0	0
18	B	7	85	4	4	0	0
19	C	21	7	53	18	1	0
20	B	9	76	7	8	0	0
21	D	3	1	2	94	0	0
22	A	68	16	7	9	1	0
23	D	10	33	14	42	1	0
24	A	45	20	23	11	1	0
25	B	30	56	6	8	0	0

2023 Higher Chemistry Section 2 Marks Data Analysis

2023 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.7	65.1%	46	1.2%
Q1aai	1	0.4	40.4%	28	1.1%
Q1aiiiA	2	0.8	40.5%	45	1.1%
Q1aiiiB	1	0.5	52.0%	49	3.8%
Q1bi	1	0.5	48.9%	44	2.2%
Q1bii	2	0.8	38.1%	42	1.4%
Q1biii	1	0.5	53.8%	30	1.2%
Q2ai	3	1.8	60.9%	57	3.4%
Q2aai	1	0.9	87.5%	32	0.7%
Q2bi	2	1.6	81.6%	50	4.1%
Q2bii	2	1.5	75.2%	55	1.4%
Q2biii	3	0.5	17.1%	46	4.0%
Q3aiA	1	0.7	72.3%	51	4.1%
Q3aiB	1	0.7	72.0%	39	2.1%
Q3aiiA	1	0.8	77.2%	50	1.3%
Q3aiiB	1	0.6	57.0%	52	2.3%
Q3aiiC	1	0.6	64.2%	46	1.0%
Q3aiiD	1	0.8	78.3%	30	0.8%
Q3aiii	1	0.3	25.7%	50	6.3%
Q3aiivA	1	0.8	77.5%	32	1.9%
Q3aiivB	1	0.5	45.3%	36	8.4%
Q3bi	1	0.9	91.0%	39	1.8%
Q3bii	1	0.3	34.1%	53	5.3%
Q3biii	1	0.8	84.4%	33	3.6%
Q3c	1	0.1	12.7%	26	2.9%
Q3d	1	0.3	29.2%	39	11.0%
Q4	3	1.2	39.6%	65	10.5%
Q5aiA	3	1.5	51.0%	62	4.0%
Q5aiB	2	1.3	66.7%	54	4.4%
Q5aai	2	1.9	93.2%	24	0.9%
Q5bi	2	0.4	21.0%	41	9.9%
Q5biiA	1	0.5	49.5%	37	3.6%
Q5biiB	1	0.8	77.3%	49	1.2%
Q5biiC	1	0.2	17.8%	44	2.2%
Q5c	2	1.7	84.1%	34	1.6%
Q6aiA	1	0.6	63.3%	23	25.7%
Q6aiB	1	0.8	82.7%	45	3.4%
Q6aai	2	1.1	54.2%	58	5.4%
Q6aiii	1	0.2	16.0%	30	2.7%
Q6biA	1	0.1	6.5%	23	3.9%
Q6biB	1	0.5	49.6%	35	1.0%
Q6bii	1	0.8	81.0%	36	0.8%
Q6ci	1	0.5	45.9%	46	3.6%
Q6ciiA	2	0.6	31.1%	48	4.5%
Q6ciiB	1	0.6	58.6%	24	3.7%
Q6di	1	0.2	17.4%	33	5.1%
Q6dii	1	0.3	28.1%	24	3.4%
Q7ai	1	0.8	76.3%	37	2.5%
Q7aai	1	0.7	65.5%	25	2.0%
Q7aiii	3	1.6	53.2%	59	8.7%
Q7b	1	0.6	56.8%	36	2.3%
Q7ci	2	0.7	36.9%	43	2.6%
Q7cii	2	1.4	71.5%	49	3.8%
Q8	3	1.4	46.7%	64	8.0%
Q9ai	2	1	48.7%	43	1.7%
Q9aai	1	0.2	19.1%	26	1.2%
Q9bi	1	0.7	70.2%	40	1.6%
Q9bii	1	0.8	75.2%	48	3.0%
Q9ci	1	0.5	54.7%	50	4.5%
Q9cii	1	0.5	46.0%	51	8.9%
Q9d	1	0.4	41.6%	28	2.4%
Q9e	1	0.2	21.0%	35	3.3%
Q10a	2	0.7	36.8%	34	9.5%
Q10bi	1	0.6	59.7%	27	2.0%
Q10bii	1	0.4	41.8%	37	3.2%
Q10c	2	1.2	58.7%	53	8.1%
Q10di	1	0.4	37.4%	43	8.4%
Q10dii	1	0.6	55.4%	44	5.6%
Paper	95	49.7	52.3%		

2023 Higher Chemistry Section 2 Frequency of Marks Analysis

2023 Item	0	1	2	3
Q1ai	35	65		
Q1aii	60	40		
Q1aiiiA	39	42	20	
Q1aiiiB	48	52		
Q1bi	51	49		
Q1bii	39	46	15	
Q1biii	46	54		
Q2ai	22	15	22	41
Q2aii	12	88		
Q2bi	11	14	75	
Q2bii	22	6	72	
Q2biii	70	18	4	9
Q3aiA	28	72		
Q3aiB	28	72		
Q3aiiA	23	77		
Q3aiiB	43	57		
Q3aiiC	36	64		
Q3aiiD	22	78		
Q3aiii	74	26		
Q3aivA	22	78		
Q3aivB	55	45		
Q3bi	9	91		
Q3bii	66	34		
Q3biii	16	84		
Q3c	87	13		
Q3d	71	29		
Q4	31	33	23	13
Q5aiA	32	16	18	34
Q5aiB	29	10	62	
Q5aii	3	8	89	
Q5bi	74	9	16	
Q5biiA	51	49		
Q5biiB	23	77		
Q5biiC	82	18		
Q5c	8	16	76	
Q6aiA	37	63		
Q6aiB	17	83		
Q6aii	29	34	37	
Q6aiii	84	16		
Q6biA	94	6		
Q6biB	50	50		
Q6bii	19	81		
Q6ci	54	46		
Q6ciiA	39	60	1	
Q6ciiB	41	59		
Q6di	83	17		
Q6dii	72	28		
Q7ai	24	76		
Q7aii	34	66		
Q7aiii	26	26	9	39
Q7b	43	57		
Q7ci	34	57	8	
Q7cii	13	31	56	
Q8	19	36	30	15
Q9ai	31	41	28	
Q9aii	81	19		
Q9bi	30	70		
Q9bii	25	75		
Q9ci	45	55		
Q9cii	54	46		
Q9d	58	42		
Q9e	79	21		
Q10a	53	21	26	
Q10bi	40	60		
Q10bii	58	42		
Q10c	37	9	54	
Q10di	63	37		
Q10dii	45	55		